Lesson 2 · Reading numbers: units and scales · for your pupils

Read the thermometer right

40 minutes · Ages 10–16 · adapt for other ages below · No electricity needed · No devices needed · Photocopy-friendly

Big question
How do we make a number mean something?

Pupils will be able to…

- Read a thermometer scale at eye level, working out what each small step is worth.
- Record a reading with its number, unit, time and place.
- Explain why two readings must be taken the same way before we compare them.

You need

- A classroom thermometer, or a paper thermometer with a sliding strip
- A clear bottle of water with a paper scale taped on
- Board and chalk; a ruler to draw a big scale
- Worksheet: Reading detective (one per pair, optional)

Before class

Hang the thermometer in the shade at pupils' eye height an hour before class. Draw a big thermometer scale on the board from 0 to 50, with five steps between each labelled ten. Make a paper thermometer if the school has none.

Time	What you do	What you can say	Pupils
Warm-up 0–5 min	Write "30" on the board. Ask: "Is 30 a lot?" Collect answers — 30 pupils, 30 minutes, 30 degrees? Then tell the story of two pupils who read 30 and 86 on one thermometer.	*"A number alone is only half the story. What else do we need to know?"*	Suggest what "30" could mean, then guess why both pupils were right.
Teach 5–13 min	Write the four parts of a reading on the board: number, unit, time, place. Teach °C, mm and %. On the board scale, show how to find what each small step is worth.	*"Count the steps between two numbers. Ten degrees, five steps — so each step is two degrees."*	Copy the four parts; work out the step size on two scales.
Eye level 13–20 min	Hold up the bottle with its paper scale. Pupils look at the water level from above, from below and level, and say what they notice. Agree the rule: eye level with the mark.	*"Bend or stand until your eye is level with the water. Now read it."*	Compare readings from three angles and explain the difference.
Reading round 20–34 min	Pairs read the board scale as you move a chalk line, writing full readings. Then they read the real or paper thermometer. If you have a real one, put it in the sun for five minutes and read again.	*"Is it fair to compare the sun reading with the shade reading? What changed besides the weather?"*	Write three full readings and explain one unfair comparison.
Wrap-up 34–40 min	Set up a class log by the door with columns for date, time, place and °C. Exit ticket: each pupil writes one full reading.	*"From tomorrow, one pupil a day reads the thermometer at the same time, in the same place."*	Complete the exit ticket; volunteer for the reading rota.

Check understanding
Listen for full readings — "27 °C, 10 am, by the door" — not just "27". Watch that pupils bend to eye level. If someone says the sun reading proves it got hotter, ask: "What else changed?"

Exit ticket
Write one full reading: number, unit, time and place.

Common mix-ups

Pupils may think...	Help them see...
The unit is extra — the number is what counts.	Write 30 on the board, then 30 °C and 30 °F. Same number, very different temperatures — the unit changes everything.
Any angle is fine if you look carefully.	Show the bottle from above and below: the level seems to move against the scale. Only eye level gives the true reading.
A higher number always means a hotter day.	Only if the readings were taken the same way. A thermometer in the sun or read in the afternoon shows more anyway.

Adapt it

Younger pupils
Ages 6–9: use a paper thermometer with a sliding red strip. Pupils move it to "hot" and "cold" and read it with their eyes level with the top.

Older pupils and university
Pupils convert between °C and °F using °F = °C × 9 ÷ 5 + 32, then plan a fair test: is it warmer by the window or the door?

Large classes
Draw the scale big on the board. Move a chalk line and have rows call out the reading together; pick one row to add the unit, time and place.

No board or no space
Tape a paper scale to a wall or tree at pupils' eye height. Slide a paper strip up and down; pupils read it at eye level and say the unit.

Short on time? Tell the 30-and-86 story, teach the four parts of a reading, and read the board scale together (20 minutes).

What you need to know

A number on its own means little. Give every reading a unit, read the scale at eye level, note the time and place — and compare only readings taken the same way.

Unit
The name that tells you what a number measures, such as °C for temperature or mm for rainfall. A number without a unit is only half a reading.

Scale
The marks and numbers on an instrument that you read against, like the lines on a ruler or a thermometer.

Eye level
Looking straight across at the mark, not from above or below. Looking at an angle makes the reading seem higher or lower.

Fair comparison
Comparing readings taken the same way — same unit, same place, same time of day — so a difference tells you something real.

My notes — how did it go?